

常微分作业 (7)

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1.

$$1) A(t)A(s) = \begin{bmatrix} 2t & 0 \\ 0 & 2s \end{bmatrix}, A(s)A(t) = \begin{bmatrix} 2s & 0 \\ 0 & 2t \end{bmatrix}$$

故 $t \neq s$ 时, $A(t)A(s) \neq A(s)A(t)$

2)

$$\begin{aligned} & \left(\int_0^t A(\tau) d\tau \right)^n \\ &= \int_0^t \int_0^t \cdots \int_0^t A(t_1)A(t_2) \cdots A(t_n) dt_n \cdots dt_2 dt_1 \\ &= \sum_{\sigma \in S_n} \int_0^t \int_0^{t_{\sigma(1)}} \cdots \int_0^{t_{\sigma(n-1)}} A(t_1)A(t_2) \cdots A(t_n) dt_{\sigma(n)} \cdots dt_{\sigma(2)} dt_{\sigma(1)} \\ &= \sum_{\sigma \in S_n} \int_0^t \int_0^{t_1} \cdots \int_0^{t_{n-1}} A(t_{\sigma(1)})A(t_{\sigma(2)}) \cdots A(t_{\sigma(n)}) dt_n \cdots dt_2 dt_1 \end{aligned}$$

第二个等式是根据 t_i 的大小关系, 排序得到的。

根据第一问, $t \neq s$ 时, $A(t)A(s) \neq A(s)A(t)$, 从而对于两个不同的置换 σ, τ , 有

$$A(t_{\sigma(1)})A(t_{\sigma(2)}) \cdots A(t_{\sigma(n)}) \neq A(t_{\tau(1)})A(t_{\tau(2)}) \cdots A(t_{\tau(n)})$$

从而

$$\left(\int_0^t A(\tau) d\tau \right)^n \neq n! \sum_{\sigma \in S_n} \int_0^t \int_0^{t_1} \cdots \int_0^{t_{n-1}} A(t_1)A(t_2) \cdots A(t_n) dt_n \cdots dt_2 dt_1$$

即

$$B_n(t) \neq \frac{\left(\int_0^t A(\tau) d\tau \right)}{n!}$$

$$3) \int_0^t A(\tau) d\tau = \begin{bmatrix} 0 & t^2 \\ t & 0 \end{bmatrix}, D(t) = e^{\int_0^t A(\tau) d\tau} = \begin{bmatrix} \frac{e^{t^{\frac{3}{2}}} + e^{-t^{\frac{3}{2}}}}{2} & t^{\frac{1}{2}} \frac{e^{t^{\frac{3}{2}}} - e^{-t^{\frac{3}{2}}}}{2} \\ t^{-\frac{1}{2}} \frac{e^{t^{\frac{3}{2}}} - e^{-t^{\frac{3}{2}}}}{2} & \frac{e^{t^{\frac{3}{2}}} + e^{-t^{\frac{3}{2}}}}{2} \end{bmatrix}$$

计算 $D'(t)$ 中 $(1,1)$ 位置的元素, 为 $3t^{\frac{1}{2}} \frac{e^{t^{\frac{3}{2}}} - e^{-t^{\frac{3}{2}}}}{4} A(t)D(t)$ 中 $(1,1)$ 位置的元素为 $t^{\frac{1}{2}} \frac{e^{t^{\frac{3}{2}}} - e^{-t^{\frac{3}{2}}}}{2}$, 二者不相等, 从而

$$D'(t) \neq A(t)D(t)$$

4)

$$X_1(t) = z + \int_0^t A(\tau) z d\tau = (I + B_1(t))z$$

$$X_2(t) = z + \int_0^t A(\tau) X_1(\tau) z d\tau = (I + B_1(t) + B_2(t))z$$

由归纳法不妨设 $X_n(t) = \sum_{i=0}^n B_i(t)z$ 则 $X_{n+1}(t) = z + \int_0^t A(\tau) X_n(\tau) z d\tau =$

$\sum_{i=0}^{n+1} B_i(t)z$, 故假设成立。从而 $X(t) = \sum_{i=0}^{\infty} B_i(t)z$

2.

1)

$$\begin{aligned} & \left(\int_s^t A(\tau) d\tau \right)^n \\ &= \int_s^t \int_s^t \cdots \int_s^t A(t_1) A(t_2) \cdots A(t_n) dt_n \cdots dt_2 dt_1 \\ &= \sum_{\sigma \in S_n} \int_s^t \int_s^{t_{\sigma(1)}} \cdots \int_s^{t_{\sigma(n-1)}} A(t_1) A(t_2) \cdots A(t_n) dt_{\sigma(n)} \cdots dt_{\sigma(2)} dt_{\sigma(1)} \\ &= \sum_{\sigma \in S_n} \int_s^t \int_s^{t_1} \cdots \int_s^{t_{n-1}} A(t_1) A(t_2) \cdots A(t_n) dt_n \cdots dt_2 dt_1 \\ &= n! \int_s^t \int_s^{t_1} \cdots \int_s^{t_{n-1}} A(t_1) A(t_2) \cdots A(t_n) dt_n \cdots dt_2 dt_1 \end{aligned}$$

从而有 $B_n(s, t) = \frac{(\int_s^t A(\tau) d\tau)^n}{n!}$

2) 类似第一题 4), 我们有

$$\Pi(t, s) = \sum_{i=0}^{\infty} B_i(s, t) = \sum_{i=0}^{\infty} \frac{(\int_s^t A(\tau) d\tau)^i}{i!} = e^{\int_s^t A(\tau) d\tau}$$

从而

$$\frac{\partial \Pi}{\partial t}(t, s) = \frac{\partial}{\partial t} \sum_{i=s}^{\infty} \frac{(\int_s^t A(\tau) d\tau)^n}{n!} = A(t) \sum_{i=1}^{\infty} \frac{(\int_0^t A(\tau) d\tau)^{n-1}}{(n-1)!} = A(t) \Pi(t, s)$$

根据 A 的交换性, 还有

$$\frac{\partial \Pi}{\partial t}(t, s) = A(t) \sum_{i=1}^{\infty} \frac{(\int_s^t A(\tau) d\tau)^{n-1}}{(n-1)!} = \sum_{i=1}^{\infty} \frac{(\int_s^t A(\tau) d\tau)^{n-1}}{(n-1)!} A(t) = \Pi(t, s) A(t)$$

3) $A(t) = \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}$ 时, $\int_0^t A(\tau) d\tau = \begin{bmatrix} \sin t & 1 - \cos t \\ \cos t - 1 & \sin t \end{bmatrix}$, 不难验证有 $A(t)A(s) = A(s)A(t)$, 从而根据上述结论

$$X(t) = e^{\int_0^t A(\tau) d\tau} z = e^{\sin t} \begin{bmatrix} \cos(1 - \cos t) & \sin(1 - \cos t) \\ -\sin(1 - \cos t) & \cos(1 - \cos t) \end{bmatrix} z$$

$$A(t) = \begin{bmatrix} 1 & t & \frac{t^2}{2} \\ & 1 & t \\ & & 1 \end{bmatrix} \text{ 时, } \int_0^t A(\tau) d\tau = \begin{bmatrix} t & \frac{t^2}{2} & \frac{t^3}{6} \\ & t & \frac{t^2}{2} \\ & & t \end{bmatrix}, \text{ 不难验证有 } A(t)A(s) =$$

$A(s)A(t)$, 从而根据上述结论

$$X(t) = e^{\int_0^t A(\tau) d\tau} z = e^t \begin{bmatrix} 1 & \frac{t^2}{2} & \frac{t^3}{6} + \frac{t^4}{8} \\ & 1 & \frac{t^2}{2} \\ & & 1 \end{bmatrix} z$$

3.

1)

$$\frac{\partial B_n}{\partial s}(t, s) = -A(s) \frac{(\int_s^t A(\tau) d\tau)^{n-1}}{(n-1)!}$$

2)

$$\frac{\partial \Pi}{\partial s}(t, s) = \sum_{i=0}^{\infty} \frac{\partial B_n}{\partial s}(t, s) = \sum_{i=1}^{\infty} \frac{(\int_s^t A(\tau) d\tau)^{n-1}}{(n-1)!} - A(s) = -\Pi(t, s) A(s)$$

根据第二题和此问, 由于 $\Pi(t, s), A$ 均连续, 从而有 Π 关于 t, s 都光滑, 于是 $\Pi \in (C^1 \times C^1; M_d(\mathbb{R}))$

$$3) (\det \Pi(t, s))' = \text{tr}(\Pi(t, s)^{-1} \frac{d\Pi(t, s)}{ds}) \det(\Pi(t, s)) = -\text{tr}(A) \det(\Pi(t, s)),$$

从而 η 满足 $\dot{x} = -\text{tr} A(s)x$

从而 $\eta(s) = e^{\int_s^t \text{tr}(A)(\tau) d\tau}$

$$4. \text{ 设 } \phi_1(t) = \begin{cases} 0, & x \leq 0, \\ (t, 0, 0, \dots, 0), & x > 0 \end{cases}, \phi_2(t) = \begin{cases} 0, & x \geq 0, \\ (0, t, 0, \dots, 0), & x < 0 \end{cases}$$

$2 < t \leq d, \phi_i(t) = (0, \dots, 0, t, 0, \dots, 0)$, 即第 i 个分量为 0. 不难验证

这 d 个函数线性无关且

$$W(\phi_1(t), \phi_2(t), \dots, \phi_d(t)) \equiv 0$$